

SHD-CCP Protocol: Mathematical Proof of Stability

Subject: Verification of Transmission Specifications and Latency Targets \

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1. Structural Stability Proof (Packet Size Integrity)

Objective: Prove that the defined packet structures are mathematically consistent with the cryptographic requirements of the Falcon algorithm and the LPF payload.

1.1 Variable Definitions

Let S_{LPF} be the size of the Logical Packet Frame. Let S_{PK_512} and S_{PK_1024} be the sizes of the Falcon Public Keys. Let S_{SIG_512} and S_{SIG_1024} be the sizes of the Falcon Signatures.

Constants (Falcon Spec Round 3):

- $S_{LPF} = 16$ Bytes (Fixed 128-bit frame)
- $S_{PK_512} = 897$ Bytes
- $S_{SIG_512} = 666$ Bytes
- $S_{PK_1024} = 1793$ Bytes
- $S_{SIG_1024} = 1280$ Bytes

1.2 Genesis Packet Derivation (Type G)

The Genesis packet (P_G) must contain the payload, the public key for the chain, and a self-signed signature.

Formula: $Size(P_G) = S_{LPF} + S_{PK} + S_{SIG}$

- **Proof for Falcon-512:**

$$16 + 897 + 666 = 1,579 \text{ Bytes}$$

Status: MATCHES CALIBRATION SHEET

- **Proof for Falcon-1024:**

$$16 + 1,793 + 1,280 = 3,089 \text{ Bytes}$$

Status: MATCHES CALIBRATION SHEET

1.3 Chained Packet Derivation (Type C)

The Chained packet (P_C) contains the payload, the signature of the previous packet ($Linkage$), and its own signature.

Formula: $Size(P_C) = S_{LPF} + S_{Linkage} + S_{SIG}$ Where $S_{Linkage} = S_{SIG}$ (Hash Pointer)

- **Proof for Falcon-512:**

$$16 + 666 + 666 = 1,348 \text{ Bytes}$$

Status: MATCHES CALIBRATION SHEET

- **Proof for Falcon-1024:**

$$16 + 1,280 + 1,280 = 2,576 \text{ Bytes}$$

Status: MATCHES CALIBRATION SHEET

2. Temporal Stability Proof (Latency Derivation)

Objective: Proof that the protocol meets the "< 10ms" core latency target across standard network conditions.

2.1 Physics of Transmission

Transmission latency (L_t) is defined as the time required to push all bits of the packet onto the wire.

Formula: $L_t = \frac{Size_{bits}}{Bandwidth_{bps}}$

2.2 Bit Conversions

- **Genesis-512:** $1,579 \text{ Bytes} \times 8 = 12,632 \text{ bits}$
- **Chained-512:** $1,348 \text{ Bytes} \times 8 = 10,784 \text{ bits}$
- **Genesis-1024:** $3,089 \text{ Bytes} \times 8 = 24,712 \text{ bits}$
- **Chained-1024:** $2,576 \text{ Bytes} \times 8 = 20,608 \text{ bits}$

2.3 Verification Calculations

Scenario A: 1 Gigabit Network ($1 \times 10^9 \text{ bps}$)

- **Genesis-512:** $\frac{12,632}{10^9} = 0.000012632 \text{ s} \approx \mathbf{12.63\mu s}$
- **Chained-512:** $\frac{10,784}{10^9} = 0.000010784 \text{ s} \approx \mathbf{10.78\mu s}$
- **Chained-1024:** $\frac{20,608}{10^9} = 0.000020608 \text{ s} \approx \mathbf{20.61\mu s}$

- *Conclusion: Validates "μs range" claims.*

Scenario B: 100 Megabit Network (1×10^8 bps)

- **Genesis-512:** $\frac{12,632}{10^8} = 0.00012632 \text{ s} \approx \mathbf{126.3\mu s}$
- *Conclusion: Validates 126 μs benchmark.*

Scenario C: 10 Megabit Network (1×10^7 bps)

- **Genesis-512:** $\frac{12,632}{10^7} = 0.0012632 \text{ s} \approx \mathbf{1.26ms}$
- *Conclusion: Validates 1.26 ms benchmark. Even at low speeds, latency is well below the < 10ms target.*

3. Overhead & Efficiency Analysis

Objective: Determine the "Security Tax" — the ratio of security metadata to actual payload.

Formula: $Ratio = \frac{TotalSize}{PayloadSize}$

- **Chained-512 Efficiency:**

$$1,348/16 = 84.25$$

Interpretation: For every 1 byte of neural-symbolic data, 84.25 bytes of cryptographic proof are transmitted.

- **Chained-1024 Efficiency:**

$$2,576/16 = 161.0$$

Interpretation: Military-grade security doubles the bandwidth cost.

Stability Conclusion: While the overhead ratio is high (~84x), the absolute size (1.3KB) is negligible for modern networks. The protocol is **stable** because even at high overhead ratios, the raw throughput (Packets Per Second) remains immense:

- @ 1 Gbps (Chained-512): $\approx 92,730$ packets/sec
- This supports nearly **100,000 neural-symbolic transactions per second**, far exceeding the inference speed of current LLMs.